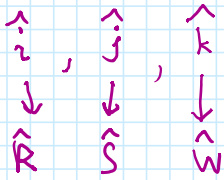
Relative motion

Thursday, October 1, 2026 9:54 AM

1) General derivation and circular reference orbits (Curtis chapter 7)
2) Curvilinear coordinates, elliptical reference orbits



Reference orbit - primary satellite
Secondary

Radial along-track cross-track

$$\delta\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

Relative orbital elements:

- Relative eccentricity vector
- Relative inclination vector

δi = angle between the orbital planes

$$\sin(\delta i)\cdot\begin{bmatrix}\cos\theta\\ \sin\theta\end{bmatrix} \qquad \theta$$

Argument of latitude of the relative ascending node (the point where the second sc crosses the orbital plane of the first going in the +k direction)

Linearization: derivation of the Hill-Clohessy-Wiltshire (HCW) equations

$$\ddot{\vec{r}}_A = -\frac{\mu}{r_A^3}\vec{r}_A$$

$$\ddot{\vec{r}}_B = -\frac{\mu}{r_B^3}\vec{r}_B + \vec{F}_{thrust}$$

$$\delta\vec{r} = \vec{r}_B - \vec{r}_A$$

$$\delta\ddot{\vec{r}} = -\frac{\mu}{r_B^3}\vec{r}_B + \vec{F}_{th} + \frac{\mu}{r_A^3}\vec{r}_A$$

Assumption:

$$\delta r \ll r_B, r_A$$

$$\vec{r}_B = \vec{r}_A + \delta\vec{r}$$

$$\frac{\vec{r}_B}{r_B^3} = \frac{\vec{r}_A + \delta\vec{r}}{\left(r_A^2 + 2\vec{r}_A\cdot\delta\vec{r} + \delta r^2\right)^{3/2}}$$

$$= \frac{\vec{r}_A + \delta\vec{r}}{r_A^3}\left(1 + \frac{2\vec{r}_A\cdot\delta\vec{r}}{r_A^2} + \dots\right)^{-\frac{3}{2}}$$

$$(1+x)^n = 1 + nx + \dots$$

$$\frac{\vec{r}_B}{r_B^3} \approx \frac{\vec{r}_A + \delta\vec{r}}{r_A^3}\left(1 - \frac{3\vec{r}_A\cdot\delta\vec{r}}{r_A^2}\right) + \vec{T}_{thrust} + \frac{\mu}{r_A^3}\vec{r}_A$$

$$\delta\ddot{\vec{r}} = -\frac{\mu}{r_A^3}\left(-\hat{r}_A\frac{3\vec{r}_A\cdot\delta\vec{r}}{r_A} + \delta\vec{r}\right) + \vec{F}_{thrust}$$

Components of $\delta\vec{r} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$$\delta\ddot{\vec{r}}_{inertial} = -\frac{\mu}{r_A^3}\left(-2x\,\hat{i} + y\,\hat{j} + z\,\hat{k}\right) + \vec{F}_{thrust}$$

$$\delta\ddot{\vec{r}} = \delta\ddot{\vec{r}}_{inertial} - \underbrace{\dot{\vec{\Omega}}\times\delta\vec{r}}_{\text{tangential acc}} - \underbrace{2\vec{\Omega}\times\delta\dot{\vec{r}}}_{\text{Coriolis}} - \underbrace{\vec{\Omega}\times(\vec{\Omega}\times\delta\vec{r})}_{\text{centrifugal}}$$

$\vec{\Omega}$ = Orbital angular velocity of the primary $= n\hat{k}$ $\quad n = \sqrt{\frac{\mu}{r_A^3}}$

For circular motion, this is a constant equal to the mean motion times k unit vector

Tangential acceleration is then 0

For the other two terms, expand in components

$$\vec{\Omega}\times\delta\vec{r} = n\left(-\dot{y}\,\hat{i} + \dot{x}\,\hat{j}\right)$$

$$\vec{\Omega}\times\left(\vec{\Omega}\times\delta\vec{r}\right) = -n^2\left(x\,\hat{i} + y\,\hat{j}\right)$$

$$-\frac{\mu}{r_A^3}\left(-2x\,\hat{i} + \dots\right) - n^2\left(x\,\hat{i} + \dots\right)$$

$$-\frac{\mu}{r_A^2}(-3x\hat{i}+\delta\vec{r}) = -n^2\left(-2x\hat{i}+y\hat{j}+z\hat{k}\right)$$

$$\delta\ddot{\vec{r}} = -n^2(-2x\hat{i}+y\hat{j}+z\hat{k}) - 2n(-\dot{y}\hat{i}+\dot{x}\hat{j}) + n^2(x\hat{i}+y\hat{j}) + \vec{F}_{th}$$

Hill-Clohessy-Wiltshire (HCW) equations

$$\ddot{x} - 2n\dot{y} - 3n^2x = f_x$$
$$\ddot{y} + 2n\dot{x} = f_y$$
$$\ddot{z} + n^2z = f_z$$

Lecture 11
10/6/2026

Corrections to lecture 8
- Signs on Kepler's equation
- Definition of beta sign error

See announcements on ELMS

Solving the HCW equations
Assume no external forces $f_x = f_y = f_z = 0$

One-dimensional z equation decoupled from the rest

$$z(t) = z_0\cos nt + \frac{\dot{z}_0}{n}\sin nt$$

Two coupled x-y equations in the orbital plane

$$x^{(3)} = 2n\ddot{y} + 3n^2\dot{x}$$
$$= -4n^2\dot{x} + 3n^2\dot{x} \quad (-2n\dot{x})$$
$$x^{(3)} + n^2\dot{x} = 0$$

v

$$\dot{v} + n^2v = 0 \Rightarrow v(t) = v_{0x}\cos nt + v_{0y}\sin nt$$

$\ddot{x} + n^2 x = 0 \quad \rightarrow \quad v(t) = v_{0x} \cos nt + v_{0y} \sin nt$

$$x(t) = \int v(t)\,dt = x_0 + \frac{v_{0x}}{n} \sin nt - \frac{v_{0y}}{n}(\cos nt - 1)$$

$$x(0) = x_0$$

$$\dot{x}(t) = v_{0x} \cos nt + v_{0y} \sin nt$$

$$\dot{x}(0) = \dot{x}_0 = v_{0x}$$

$$\ddot{x}(t) = -n v_{0x} \sin nt + n v_{0y} \cos nt$$

$$\ddot{x}(0) = \ddot{x}_0 = n v_{0y}$$

$$v_{0x} = \dot{x}_0 \qquad v_{0y} = \frac{\ddot{x}_0}{n} = 2\dot{y}_0 + 3nx_0$$

So:

$$x(t) = 4x_0 + \frac{2\dot{y}_0}{n} + \frac{\dot{x}_0}{n} \sin nt - \left(\frac{2\dot{y}_0}{n} + 3x_0\right) \cos nt$$

$$\ddot{y} = -2n\dot{x} = -2n\left(\dot{x}_0 \cos nt + \frac{\ddot{x}_0}{n} \sin nt\right)$$

$$= -2n\dot{x}_0 \cos nt - (4n\dot{y}_0 + 6n^2 x_0) \sin nt$$

$$y(t) = \frac{2\dot{x}_0}{n} \cos nt + \left(6x_0 + \frac{4\dot{y}_0}{n}\right) \sin nt - \underbrace{(6nx_0 + 3\dot{y}_0)t}$$

This secular term arises only because of the linearization approximation, which is violated given enough time.

$$\dot{y}_0 = -2nx_0$$

$$
\begin{aligned}
x(t) =& 4x_\bullet + \frac{2\dot{y}_\bullet}{n} + \frac{\dot{x}_\bullet}{n}\sin nt - \left(3x_\bullet + \frac{2\dot{y}_\bullet}{n}\right)\cos nt \\
y(t) =& \frac{2\dot{x}_\bullet}{n}\cos nt + \left(6x_\bullet + \frac{4\dot{y}_\bullet}{n}\right)\sin nt \\
& - (6nx_\bullet + 3\dot{y}_\bullet)t - \frac{2\dot{x}_\bullet}{n} + y_\bullet \\
z(t) =& z_\bullet \cos nt + \frac{\dot{z}_\bullet}{n}\sin nt.
\end{aligned}
$$

$$
\begin{aligned}
\dot{x}(t) =& \dot{x}_\bullet \cos nt + (2\dot{y}_\bullet + 3nx_\bullet)\sin nt \\
\dot{y}(t) =& -2\dot{x}_\bullet \sin nt + (6nx_\bullet + 4\dot{y}_\bullet)\cos nt \\
& - (6nx_\bullet + 3\dot{y}_\bullet) \\
\dot{z}(t) =& -z_\bullet n \sin nt + \dot{z}_\bullet \cos nt.
\end{aligned}
$$

These equations can be written as 3x3 sub-matrices of the state transition matrix:

$$
[\Phi_{rr}(t)] = \left[\begin{array}{ccc} 4-3\cos nt & 0 & 0 \\ 6(\sin nt - nt) & 1 & 0 \\ \hline 0 & 0 & \cos nt \end{array}\right] \tag{7.53a}
$$

$$
[\Phi_{rv}(t)] = \left[\begin{array}{ccc} \frac{1}{n}\sin nt & \frac{2}{n}(1-\cos nt) & 0 \\ \frac{2}{n}(\cos nt - 1) & \frac{1}{n}(4\sin nt - 3nt) & 0 \\ \hline 0 & 0 & \frac{1}{n}\sin nt \end{array}\right] \tag{7.53b}
$$

$$
[\Phi_{vr}(t)] = \left[\begin{array}{ccc} 3n\sin nt & 0 & 0 \\ 6n(\cos nt - 1) & 0 & 0 \\ \hline 0 & 0 & -n\sin nt \end{array}\right] \tag{7.53c}
$$

$$
[\Phi_{vv}(t)] = \left[\begin{array}{ccc} \cos nt & 2\sin nt & 0 \\ -2\sin nt & 4\cos nt - 3 & 0 \\ \hline 0 & 0 & \cos nt \end{array}\right] \tag{7.53d}
$$

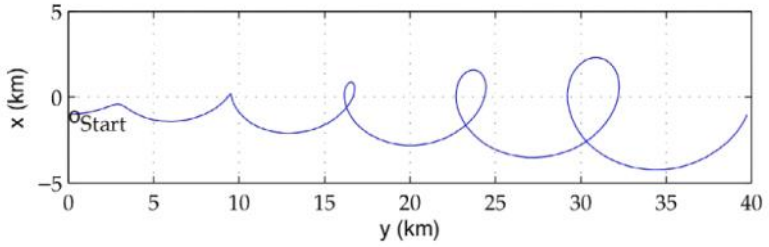


FIG. 7.6

Trajectory of *B* relative to *A* in the comoving frame during five of the target's orbits. Eccentricity of the target orbit is 0.1.

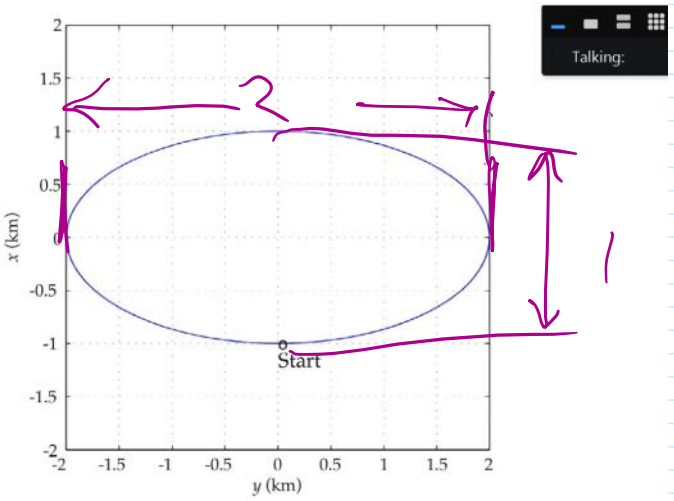


FIG. 7.7

Trajectory of *B* relative to *A* in the comoving frame during five of the target's orbits. Eccentricity of the target orbit is 0.

$e_B = 0 \qquad a_B = a_A$

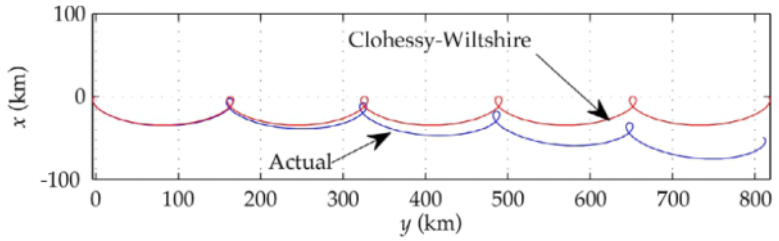


FIG. 7.8

Relative motion of a particle and its Clohessy-Wiltshire approximation.

Finding the relative coordinates from inertial state vectors for each satellite

Find RSW coordinates axes

$$\hat{R} = \frac{\vec{r}_{ref}}{|\vec{r}_{ref}|}, \quad \hat{W} = \hat{h}_{ref}, \quad \hat{S} = \hat{W} \times \hat{R}$$

Write these as a matrix, with each of these vectors being rows:

$$Q = \begin{bmatrix} \hat{R}^T \\ \hat{S}^T \\ \hat{W}^T \end{bmatrix}$$

Angular velocity of the reference frame is computed from the angular momentum

$$\vec{\Omega} = \frac{\vec{h}_{ref}}{r_{ref}^2}$$

Relative position and velocity

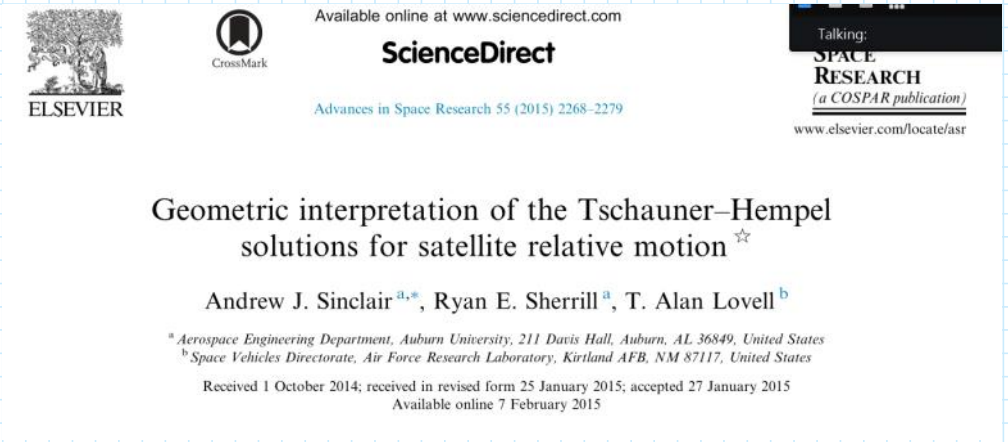
$$\delta\vec{r} = Q(\vec{r}_B - \vec{r}_{ref})$$

$$\delta\dot{\vec{r}} = Q\left[\dot{\vec{r}}_B - \dot{\vec{r}}_{ref} - \vec{\Omega} \times (\vec{r}_B - \vec{r}_{ref})\right]$$

Examples in book 7.1, plot in 7.2

Advanced techniques for relative motion

- Sullivan, Grimberg, D'Amico "Comprehensive survey and assessment of Spacecraft relative motion dynamics models, JGCD 2017 doi: 10.2514/1.G002309
- Sinclair, Sherrill, Lovell "Geometric interpretation of the Tschauner-Hempel solutions for satellite relative motion" doi: 10.1016/j.asr.2015.01.032



ELSEVIER

CrossMark

Available online at www.sciencedirect.com

ScienceDirect

Advances in Space Research 55 (2015) 2268–2279

Talking: SPACE RESEARCH (a COSPAR publication)

www.elsevier.com/locate/asr

Geometric interpretation of the Tschauner–Hempel solutions for satellite relative motion ☆

Andrew J. Sinclair [a,*], Ryan E. Sherrill [a], T. Alan Lovell [b]

[a] Aerospace Engineering Department, Auburn University, 211 Davis Hall, Auburn, AL 36849, United States
[b] Space Vehicles Directorate, Air Force Research Laboratory, Kirtland AFB, NM 87117, United States

Received 1 October 2014; received in revised form 25 January 2015; accepted 27 January 2015
Available online 7 February 2015

Trajectory is curved: use spherical coordinates instead of rectilinear

Dynamics

- Quadratic Volterra - second-order Taylor series expansion - rectilinear
- Double Transformation - nonlinear coordinate mapping of linear solutions - cylindrical or spherical coordinates

Elliptical orbits

1. Tschauner-Hempel (TH) equations
2. Yamanaka-Ankersen (YA) STM
3. Time-explicit linear models
4. Lyapunov-Floquet (LF) transformation

1. TH equations: do a variable transformation independent variable is true anomaly of the reference orbit instead of time.
2. YA STM singularity-free structurally simple STM widely used
3. Broucke, Melton models: time-explicit solutions that avoid the avoid the need to compute the true anomaly directly.
4. LF a method to relate the linear time-varying system of an elliptic orbit to the underlying LTI system (like HCW) through periodic coordinate transformations

TH/YA

$$\ddot{x} - 2\dot{f}\dot{y} - \left(\dot{f}^2 + 2\frac{\mu}{r^3}\right)x - \ddot{f}y = 0$$
$$\ddot{y} + 2\dot{f}\dot{x} + \ddot{f}x - \left(\dot{f}^2 - \frac{\mu}{r^3}\right)y = 0 \quad (1)$$
$$\ddot{z} + \frac{\mu}{r^3}z = 0$$

f = true anomaly of the ref. orbit
r = radius of the ref. orbit
$\dot{f}$ = time derivative of

Lecture 12
10/8/2026

Apply coordinate scaling and change the independent variable from time to true anomaly f

$$\tilde{\chi}' = \begin{bmatrix} \tilde{x}' \\ \tilde{y}' \\ \tilde{z}' \\ \tilde{x} \\ \tilde{y} \\ \tilde{z} \end{bmatrix}$$

Prime indicates derivative wrt true anomaly

$$\tilde{\boldsymbol{x}} = \boldsymbol{T}(f)\boldsymbol{x} \tag{2}$$

$$\boldsymbol{T}(f) = \begin{bmatrix} (1+e\cos f)\boldsymbol{I} & \boldsymbol{0} \\ -e\sin f\,\boldsymbol{I} & \frac{p^2}{h(1+e\cos f)}\boldsymbol{I} \end{bmatrix} \quad \boldsymbol{T}^{-1}(f) = \begin{bmatrix} \frac{1}{1+e\cos f}\boldsymbol{I} & \boldsymbol{0} \\ \frac{he}{p^2}\sin f\,\boldsymbol{I} & \frac{h}{p^2}(1+e\cos f)\boldsymbol{I} \end{bmatrix} \tag{3}$$

Variables e, p, h are the eccentricity, semilatus rectum, angular moment magnitude of the reference orbit.

$$\tilde{x}'' - 2\tilde{y}' - \frac{3}{1+e\cos f}\tilde{x} = 0$$
$$\tilde{y}'' + 2\tilde{x}' = 0$$
$$\tilde{z}'' + \tilde{z} = 0$$

These are Tschauner-Hempel equations

(don't normalize true anomaly because it represents elapsed time)

Yamanaka-Ankersen (2002) solution has become the standard approach to solving these equations using this integral:

$$J(f, f_0) = \int_{f_0}^{f} \frac{\mathrm{d}\zeta}{(1+e\cos\zeta)^2}$$

$$J = \frac{h}{p^2}(t-t_0) = \frac{M-M_0}{(1-e^2)^{3/2}} = \frac{E - e\sin E - E_0 + e\sin E_0}{(1-e^2)^{3/2}}$$

Solution

$$\tilde{\boldsymbol{x}} = \boldsymbol{\Psi}(f)\boldsymbol{k} \quad \boldsymbol{\Psi}(f) = [\boldsymbol{\psi}_1 \quad \boldsymbol{\psi}_2 \quad \boldsymbol{\psi}_3 \quad \boldsymbol{\psi}_4 \quad \boldsymbol{\psi}_5 \quad \boldsymbol{\psi}_6] \tag{9}$$

$$\boldsymbol{\psi}_1 = \begin{bmatrix} \sin f(1+e\cos f) \\ 2\cos f - e\sin^2 f \\ 0 \\ \cos f + e\cos 2f \\ -2\sin f(1+e\cos f) \\ 0 \end{bmatrix} \quad \boldsymbol{\psi}_2 = \begin{bmatrix} \cos f(1+e\cos f) \\ -2\sin f - e\sin f\cos f \\ 0 \\ -\sin f - e\sin 2f \\ e - 2\cos f(1+e\cos f) \\ 0 \end{bmatrix} \tag{10}$$

$$\boldsymbol{\psi}_3 = \begin{bmatrix} 1 - \frac{3e}{2} K \sin f\,(1 + e\cos f) \\ -\frac{3}{2} K (1 + e\cos f)^2 \\ 0 \\ -\frac{3e}{2} K (\cos f + e\cos 2f) - \frac{3e \sin f}{2(1+e\cos f)} \\ 3eK \sin f\,(1 + e\cos f) - \frac{3}{2} \\ 0 \end{bmatrix} \quad \boldsymbol{\psi}_4 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \tag{11}$$

$$\boldsymbol{\psi}_5 = \begin{bmatrix} 0 \\ 0 \\ \sin f \\ 0 \\ 0 \\ \cos f \end{bmatrix} \quad \boldsymbol{\psi}_6 = \begin{bmatrix} 0 \\ 0 \\ \cos f \\ 0 \\ 0 \\ -\sin f \end{bmatrix} \tag{12}$$

Replace integral J with K

$K = \frac{h(t-\tau)}{\rho^2}$ $\quad f(\tau) = 0$

$J = K - K_0$

$$\boldsymbol{\Psi}^{-1} = [\boldsymbol{\sigma}_1 \quad \boldsymbol{\sigma}_2 \quad \boldsymbol{\sigma}_3 \quad \boldsymbol{\sigma}_4 \quad \boldsymbol{\sigma}_5 \quad \boldsymbol{\sigma}_6] \tag{13}$$

$$\boldsymbol{\sigma}_1 = \begin{bmatrix} -3\frac{1+e^2+e\cos f}{(1-e^2)(1+e\cos f)} \sin f + 3K\frac{e}{1-e^2}(2+e^2+3e\cos f) \\ -3\frac{e+\cos f}{1-e^2} \\ 2\frac{2+e^2+3e\cos f}{1-e^2} \\ -3e\frac{3+e^2+2e\cos f}{(1-e^2)(1+e\cos f)} \sin f + 3K\frac{1+e^2}{1-e^2}(2+e^2+3e\cos f) \\ 0 \\ 0 \end{bmatrix} \quad \boldsymbol{\sigma}_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \tag{14}$$

$$\boldsymbol{\sigma}_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ \sin f \\ \cos f \end{bmatrix} \quad \boldsymbol{\sigma}_4 = \begin{bmatrix} \frac{\cos f\,(1+e\cos f) - 2e}{1-e^2} + 3K\frac{e^2}{1-e^2} \sin f\,(1+e\cos f) \\ -\sin f \frac{1+e\cos f}{1-e^2} \\ 2e \sin f \frac{1+e\cos f}{1-e^2} \\ 2\frac{e\cos f\,(1+e\cos f) - 1 - e^2}{1-e^2} + 3Ke\frac{1+e^2}{1-e^2} \sin f\,(1+e\cos f) \\ 0 \\ 0 \end{bmatrix} \tag{15}$$

$$\boldsymbol{\sigma}_5 = \begin{bmatrix} -\sin f \frac{2+e\cos f}{1-e^2} + 3K\frac{e}{1-e^2}(1+e\cos f)^2 \\ -\left(\frac{e+2\cos f + e\cos^2 f}{1-e^2}\right) \\ 2\frac{(1+e\cos f)^2}{1-e^2} \\ -2e \sin f \frac{2+e\cos f}{1-e^2} + 3K\frac{1+e^2}{1-e^2}(1+e\cos f)^2 \\ 0 \\ 0 \end{bmatrix} \quad \boldsymbol{\sigma}_6 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ \cos f \\ -\sin f \end{bmatrix} \tag{16}$$

State transition matrix is constructed from these matrices

STM

$$\begin{aligned} \boldsymbol{x}(t) &= \boldsymbol{T}^{-1}(f)\boldsymbol{\Psi}(f)\boldsymbol{\Psi}^{-1}(f_0)\boldsymbol{T}(f_0)\boldsymbol{x}_0 \\ &= \boldsymbol{\Omega}^{-1}(f)\boldsymbol{\Omega}(f_0)\boldsymbol{x}_0 \equiv \boldsymbol{\Phi}(f, f_0)\boldsymbol{x}_0 \end{aligned} \tag{17}$$

Time-explicit linear models
Limited to e<0.3
Broucke - relative state expressed as variations of the chief's orbital parameters
Melton - expansion in the powers of eccentricity

Lypunov-Floquet transformations
Provides periodic coordinate transformation that relates the time-varying elliptic system to a time-invariant circular system - HCW.

Method	Independent Variable	Coordinate Basis	Primary Advantage	Key Limitations / Complexity
Tschauner-Hempel (TH)	True Anomaly (θ)	Scaled Rectilinear	Fundamental linear framework for arbitrary eccentricity.	Original solutions contain singularities at perigee/apogee.
Yamanaka-Ankersen (YA) STM	True Anomaly (θ)	Rectilinear	**State-of-the-art** singularity-free analytical solution.	High computational cost due to transcendental functions and matrix inverses.
Time-Explicit (Melton/Broucke)	Time (t)	Cylindrical or Rectilinear	Avoids the overhead of solving Kepler's equation to find θ.	Accuracy often depends on series expansions in e (e.g., Melton retains $O(e^2)$).
Lyapunov-Floquet (LF)	Time (t) or True Anomaly (θ)	Virtual Circular to Elliptic Mapping	Relates complex elliptic dynamics to simple HCW intuition.	Choice of transformation (e.g., periapse vs. apoapse matching) affects local accuracy.

Orbital element representations
- Relative orbital elements
- Orbital element differences
- Geometric ROE (constants of motion of HCW)

Orbital element differences
Classical elements, take the difference between primary and secondary

$$\begin{bmatrix} \Delta a \\ \Delta e \\ \Delta i \\ \Delta \Omega \\ \Delta \omega \\ \Delta M \end{bmatrix}$$

Easy to understand, but have singularities
Alternatively: equinoctial element differences

Relative orbital elements

$$\delta a = \frac{a_s - a_p}{a_p}$$

$$\delta\lambda = v_{M_s} - v_{M_p} + (\Omega_s - \Omega_p)\cos i_p$$

Mean argument of latitude

$$u_M = M + \omega$$

Relative eccentricity vector

$$\delta\vec{e} = \begin{bmatrix} e_s \cos\omega_s - e_p \cos\omega_p \\ e_s \sin\omega_s - e_p \sin\omega_p \end{bmatrix}$$

Relative inclination vector

$$\delta\vec{i} = \begin{bmatrix} i_s - i_p \\ (\Omega_s - \Omega_p)\sin i_p \end{bmatrix}$$

These are commonly used for control of formation-flying or cluster flight spacecraft

Mapping to cartesian

$$x = a_p(\delta a - \delta e_x \cos u_p - \delta e_y \sin u_p)$$

$$y = a_p\left(-\frac{3}{2}\delta a\, u_p + \delta\lambda + 2\delta e_x \sin u_p - 2\delta e_y \cos u_p\right)$$

$$z = a_p(\delta i_x \sin u_p - \delta i_y \cos u_p)$$

$$z = a_p \left(\delta i_x \sin u_p - \delta i_y \cos u_p \right)$$

Geometric elements/parameters that define constants of the motion

- Relative ellipse

$$a_{ell} = 2\sqrt{\left(\frac{\dot{x}}{n}\right)^2 + \left[3x + \frac{2\dot{y}}{n}\right]^2}$$

- Center of the projected ellipse

$$c_{ell} = \begin{bmatrix} 4x + \frac{2\dot{y}}{n} \\ y - \frac{2\dot{x}}{n} \end{bmatrix}$$

- Relative in-plane phase

$$\beta = \arctan \frac{\dot{x}}{3nx + 2\dot{y}}$$

- Cross track amplitude and phase

$$z_{max} = \sqrt{z^2 + \left(\frac{\dot{z}}{n}\right)^2}$$

$$\psi = \arctan\left(\frac{nz}{\dot{z}}\right)$$